

DEVELOPMENT OF RAG-BASED WEB APPLICATION AND CONTRIBUTION TO CLIMADA PYTHON TOOL

INTERNSHIP REPORT



UNITED NATIONS
UNIVERSITY

UNU-EHS

Institute for Environment
and Human Security

Submitted by :

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4 Sept 2025



Organization: [United Nations University – Environment and Human Security \(UNU-EHS\)](#)

Section: [Munich Climate Insurance Initiative \(MCII\)](#)

Team: [Climate Risk Analytics Team](#)

Sub-group: AI, Remote Sensing and Modelling Group

Location: Bonn, Germany

Duration: 01 July - 29 Sept 2025 [Full-time]

Supervisor: [Dr. David Daou](#)

1. Organization Background

UNU-EHS, established in December 2003 and based in Bonn, Germany, is a research institute of the United Nations University. It serves as a think tank dedicated to strengthening human security and well-being by reducing current and future risks linked to environmental hazards and climate change. Its work is organized into two main research areas: **Risk and Adaptation**, which explores interconnected risks and climate impacts to guide science and policy, and **Transformation**, which promotes sustainable futures and drives actions that support climate resilience, equity, and human well-being (United Nations University Institute for Environment and Human Security, n.d.).

The **Munich Climate Insurance Initiative (MCII)** operates under UNU-EHS's Risk and Adaptation framework. The MCII was started as a charitable organization by experts from insurers, research institutes and NGOs in 2005 in response to the growing realization that insurance solutions can play a critical role in helping vulnerable populations adapt to climate change, as highlighted in the UN Framework Convention on Climate Change and Kyoto Protocol. MCII focuses on developing solutions for poor and vulnerable people in developing countries and communities to respond to losses and damages posed by climate change. The project office of MCII is hosted at the UNU-EHS, and is structured into three teams: Climate Policy and Governance, Climate Risk Analytics, and Climate Risk Finance Solutions.

The **Climate Risk Analytics** team aims to expand access to climate risk information for decision-makers in vulnerable countries using tools such as CLIMADA, econometrics, visualization platforms, as well as deep learning, machine learning, and remote sensing techniques.

2. Tasks

2.1. Development of the RAG-based Web Platform

Retrieval-Augmented Generation is a technique that enhances the responses of large language models (LLMs) by referring to the provided reliable external knowledge. This enables LLMs to generate content that is more accurate, contextually relevant, and domain-specific, all without needing to retrain the base model. LLMs are trained on vast datasets, but their knowledge is fixed at a certain cutoff and may become outdated or lack specificity. RAG bridges this gap by integrating up-to-date, authoritative information from outside sources, improving overall response quality while avoiding the time and cost of retraining Amazon Web Services, Inc. (n.d.).

The team was initially running the RAG (Retrieval-Augmented Generation) system using Google Colab notebooks, which is tedious and inefficient. User required to repeatedly connect

to Google Drive, read data from the drive, and install necessary packages each time, leading to frequent errors and consuming considerable time. To address these challenges, my colleague Omowonuola and I proposed a **web-based RAG application development**, providing a more streamlined, user-friendly, and reliable interface for running RAG workflows without the repetitive setup and dependency issues. The existing RAG architecture implementation was studied in detail and used as a foundation for developing the application. The task that are carried out for this project is described as follow:

a) Interface Design

The initial interface for the web-based RAG application was designed using **Figma**, allowing for a visual representation of the user experience and workflow before any coding began. The prototype included layout designs, navigation flow, and key interactive elements to ensure clarity and usability. Once the design was finalized, it was exported and submitted to the supervisor for review and feedback. After receiving approval, the frontend and backend development was proceeded, using the approved prototype as a reference to implement a consistent and user-friendly interface.

b) Frontend Development

Frontend development focuses on creating the user interface and managing user interactions within a web application. In this project, **ReactJS**, a popular JavaScript library, was used for building the frontend. A native **WebSocket** was implemented to maintain a real-time connection with the backend, using the client's user ID for identification. This allows the application to receive instant updates on asynchronous tasks, such as document indexing and question-answer processing, enhancing both responsiveness and user experience. Additionally, the frontend communicates with backend APIs using **Axios**, a JavaScript library that enables the application to send and receive data through GET, POST, PUT, DELETE, and other HTTP requests.

c) Backend Development

Backend development is a crucial aspect of web application development, as it involves building the server-side components that handle the application's core functionality, data processing, and integration with other services. In this project, the database design was first created using dbdiagram.io ([link](#)), allowing for a clear visualization of tables, relationships, and constraints before implementation.

The backend is developed using **FastAPI**, a modern, high-performance Python web framework that supports asynchronous programming, automatic data validation, and seamless integration with RESTful APIs. To handle document indexing and question answering using the RAG service, **Celery** is integrated with the backend. **Redis** is used as a message broker to facilitate communication between the backend and background task manager (Celery). When a task—such as document indexing or question-answering via the RAG service—is completed, Redis publishes a message notifying the system that the task has finished. This allows the backend to efficiently track task status, trigger subsequent actions, and provide real-time updates to users without constantly polling the server. By acting as a fast, in-memory message broker, Redis ensures smooth coordination between asynchronous tasks and the main application workflow, improving both performance and responsiveness.

For real-time communication between the client and server, **WebSockets** are implemented, enabling instant updates and bidirectional data flow. This combination of tools ensures the

backend is robust, scalable, and capable of supporting complex application logic and high-concurrency scenarios.

d) Containerization and Application Deployment

In this project, the frontend, backend, and RAG service are developed in separate repositories to maintain modularity and streamline development. For deployment, a final repository is created that integrates all three components as submodules, which are then containerized using **Docker**.

The fully containerized application is deployed on **DigitalOcean**, a cloud infrastructure platform, making it accessible to a wider audience. This deployment strategy ensures reliability, improves resource management, and enables seamless updates and expansion as the application grows.

e) Outcomes

Figure 1 represents the landing page of the application that comprises of Navbar with links to Home, About, and User page.

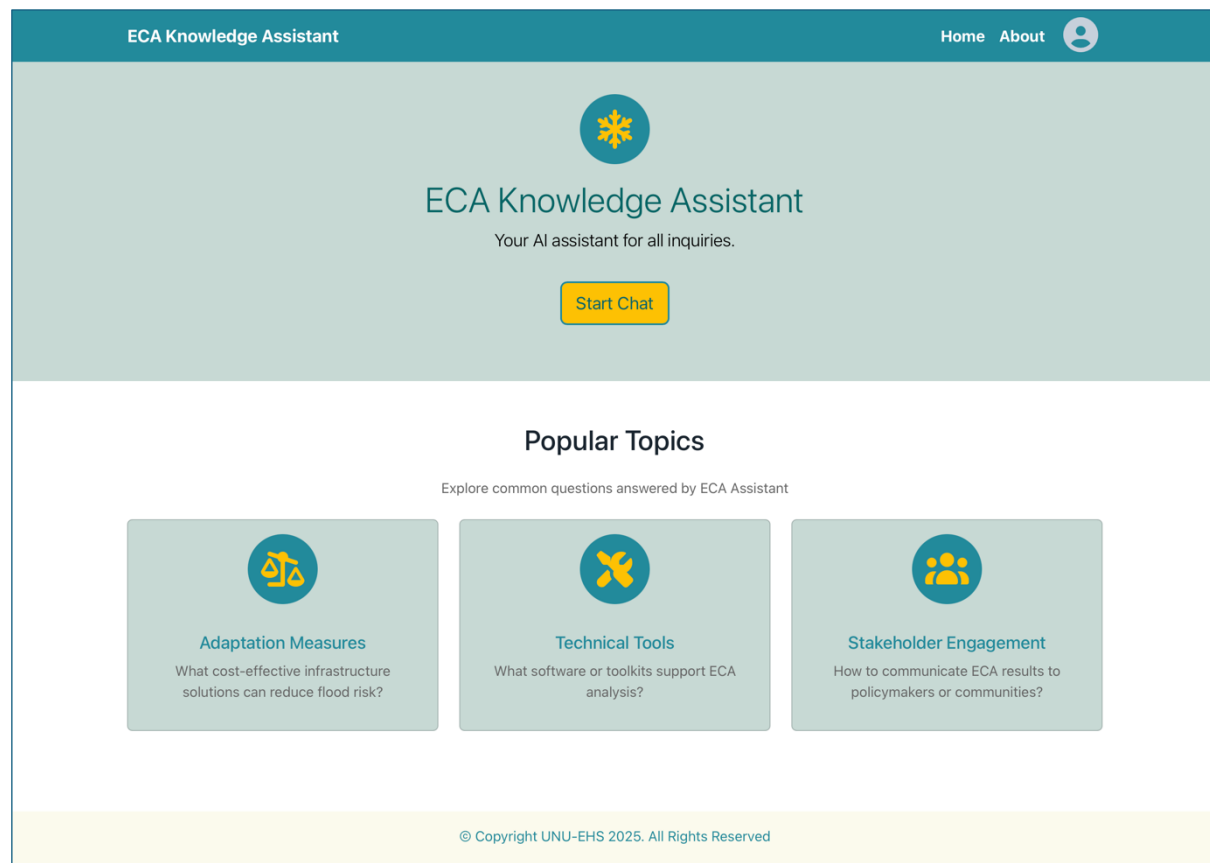


Figure 1: Home Page

Figure 2 represents the About page of the application that consists of a brief description and key features about the application, along with the team members' information who are involved in this project.

Figure 3 shows the features related to the User component of the application that consists of user login, signup, and forgot password form. After signing up for the application and email verification, the user can log in to the application.

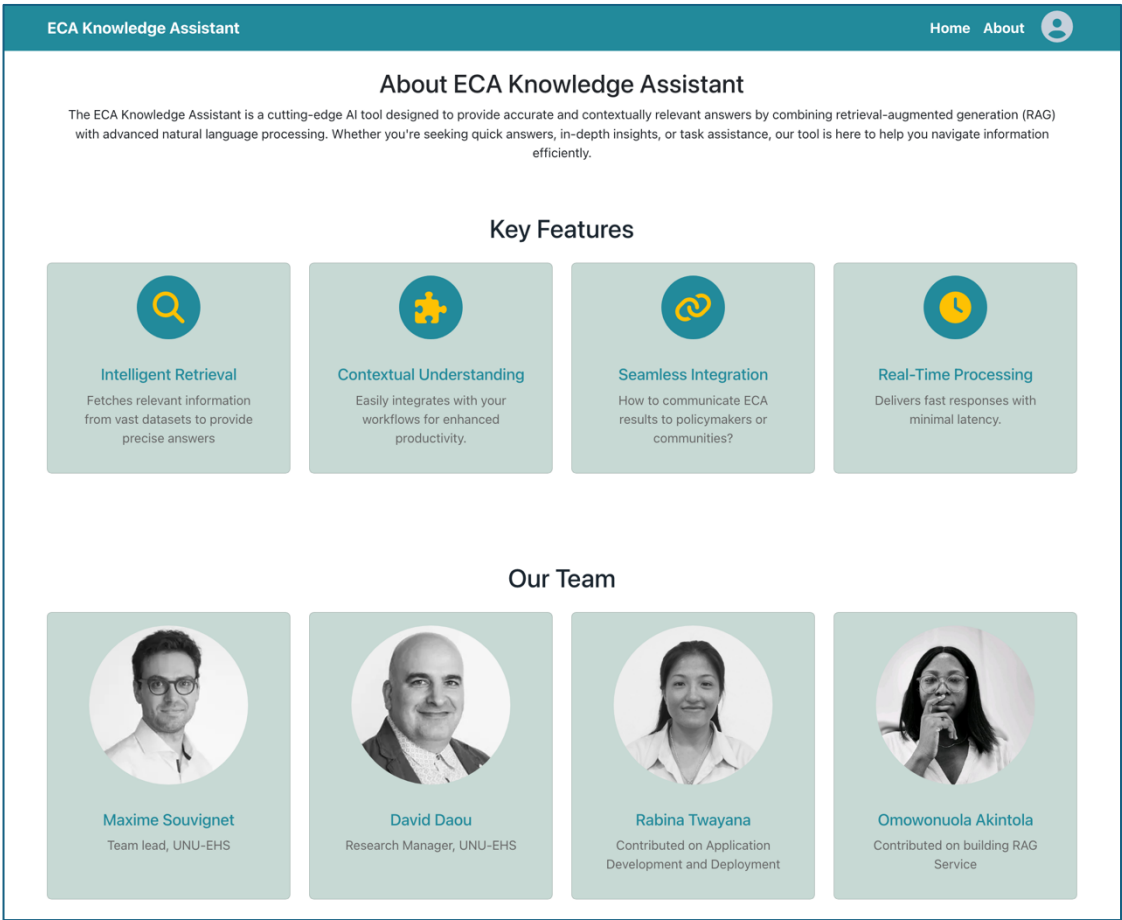


Figure 2: About Page

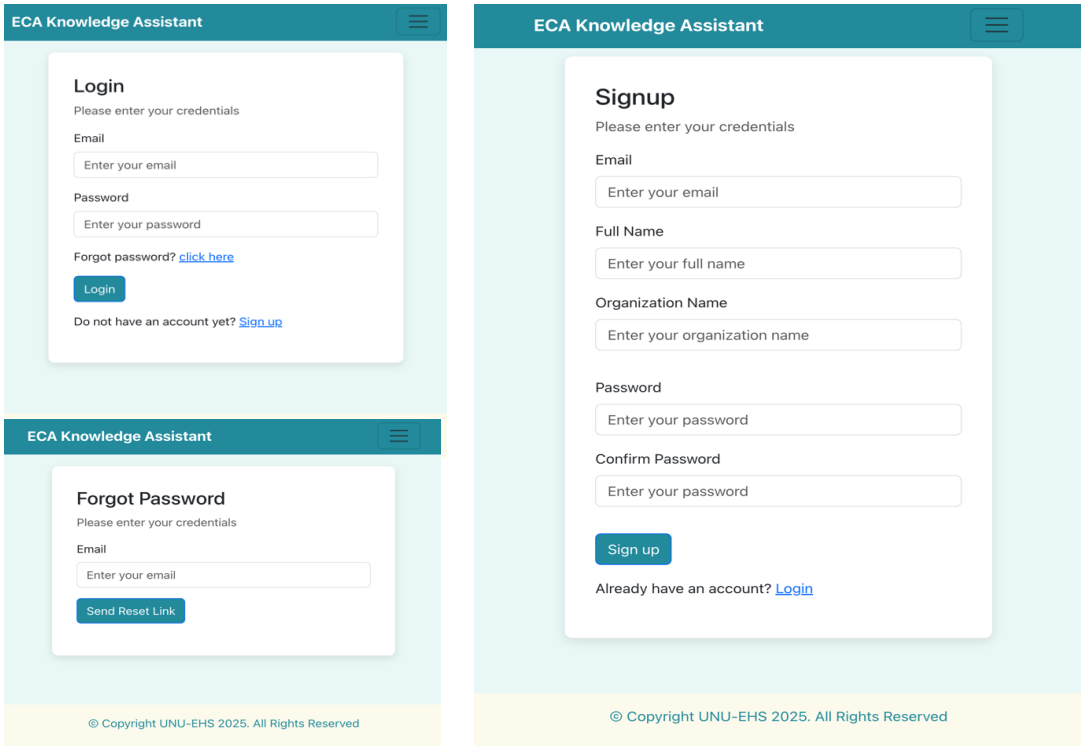


Figure 3: Signup, Login and Forgot Password Page

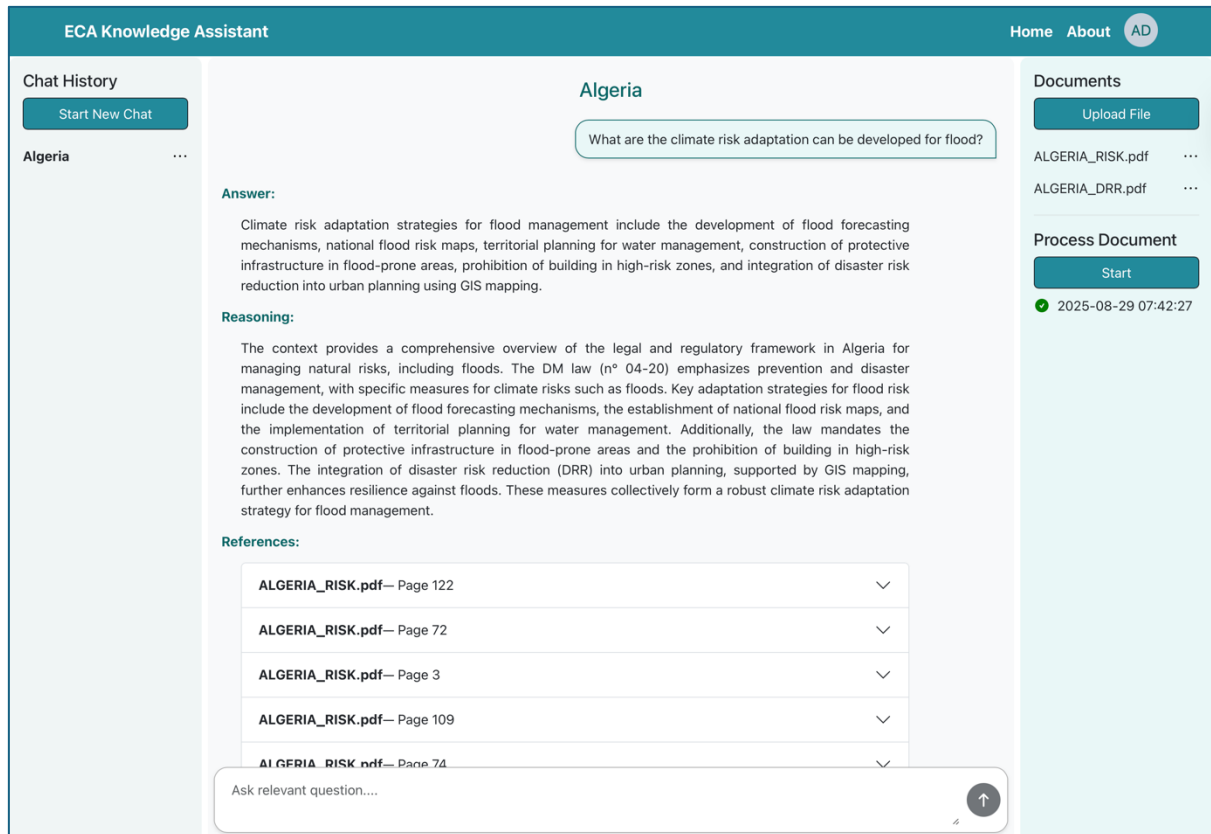


Figure 4: Main Page

Figure 4 represents the main page of the application. It has four major components: Chat history, Documents, Process Documents, and the main Chat window. Within the chat history window, the user can create/start a new chat. The user can also rename or delete the chat from the history. In the Documents section, users can upload the required files, based on which user can generate the answers. After uploading all the required documents, the user needs to process the documents by clicking the *start* button. In this step, indexing will be generated for the uploaded documents in the backend using the RAG service. After completing this step, now user can proceed with asking a question and get the answers. The answers consist of the answer to the question itself, along with the reasoning mentioning why the answer is correct, and the list of references from where the information was extracted.

Figure 5 represents the backend admin page, which is built using [SQLAlchemy admin](#). Through the admin interface, administrators can interact directly with the underlying database models, perform CRUD (Create, Read, Update, Delete) operations, and monitor system activities in a structured and user-friendly manner.

Figure 6 represents the API documentation built with FastAPI. It displays various endpoints categorized under "auth," "users," and "chats," including methods like GET, POST, PATCH, and DELETE for authentication, user management, chat operations, document handling, indexing, and Celery worker tasks.

Admin

Users

Chats

Messages

Documents

Indexings

Logout

Documents

Export

Actions

	id	deleted_at	chat_id	name	file_path	content_type
☐	1		1	ALGERIA_DRR.pdf	app/static/uploads/2ffc614819344ca9f1ea0f05b8bb01.pdf	application/pdf
☐	2		1	ALGERIA_RISK.pdf	app/static/uploads/bfe69eb389154f1f839dadb7cf4939a0.pdf	application/pdf
☐	3		2	NDC1_Algérie.pdf	app/static/uploads/77f54839fb2c41e997ceadbfcc1cb7ba.pdf	application/pdf
☐	4		2	NC3_Algeria.pdf	app/static/uploads/55936e868dd5405e962236c0b6355166.pdf	application/pdf
☐	5		2	NAP_Proposal_Algérie.pdf	app/static/uploads/c75305de065f4ebfa5b10ece5de8ffc9.pdf	application/pdf
☐	6		2	ALGERIA_SENDAI.pdf	app/static/uploads/ed0a9a539e88458084ad6bd810b5211c.pdf	application/pdf
☐	7		2	ALGERIA_RISK4.pdf	app/static/uploads/7c01939f58244ad2a9fafa2b03ad41d4.pdf	application/pdf
☐	8		2	ALGERIA_RISK3.pdf	app/static/uploads/d2f04ebb183c499bbab2e7b5f6ea77fb.pdf	application/pdf
☐	9		2	ALGERIA_RISK2.pdf	app/static/uploads/8308a8760b984eb2ba3cef9b4d2ec6cc.pdf	application/pdf
☐	10		2	ALGERIA_RISK1.pdf	app/static/uploads/db935d8af7e94159ac799036b6dc3974.pdf	application/pdf

Showing 1 to 10 of 23 items

< prev

1

2

3

Figure 6: Admin Page

ECA Assistant Application FastAPI

0.3.0 OAS 3.1

OpenAPI JSON

Authorize

auth

GET /api/v1/auth/info Auth:Info

POST /api/v1/auth/signup Auth:Signup

POST /api/v1/auth/signin Auth:Signin

POST /api/v1/auth/activate Auth:Activate

POST /api/v1/auth/forgot_password Auth:Forgot-Password

POST /api/v1/auth/reset_password Auth:Forgot-Password

users

GET /api/v1/users Users:All

PATCH /api/v1/users User:Patch-By-Id

DELETE /api/v1/users User:Delete-By-Id

GET /api/v1/users/{user_id} User:Info-By-Id

chats

GET /api/v1/chat Chat:Get-Chat-List

POST /api/v1/chat User:Create-Chat

PATCH /api/v1/chat/{chat_id} Chat:Patch-By-Id

DELETE /api/v1/chat/{chat_id} User:Delete-By-Id

GET /api/v1/chat/messages Chat:Get-Message-List-By-Chat-Id

POST /api/v1/chat/message User:Create-Chat

GET /api/v1/chat/documents Chat:Get-Chat-List

POST /api/v1/chat/document User:Create-Chat

PATCH /api/v1/chat/document/{document_id} Chat:Patch-By-Id

DELETE /api/v1/chat/document/{document_id} User:Delete-By-Id

GET /api/v1/chat/indexings Indexing:Get-Indexing-List

POST /api/v1/chat/indexing User:Create-Indexing-Using-Rag-Service

celery_worker

GET /api/v1/celery_worker/task_status/{task_id} Tasks:Get-Task-Status

Figure 5: API Docs

2.2. Contribution to CLIMADA Development

In this task, both the older version of CLIMADA (v3.3.2) and the newer version (v6.0.1) were installed in separate conda environments. Since the newer release introduced significant changes, the team aimed to upgrade from the older version to take advantage of the enhanced functionalities. However, several issues arose during the migration. The main challenge was that the hazard dataset in .matlab format, supported in the older version, was no longer compatible with the newer one. To resolve this, the dataset was converted into .hdf5 format using CLIMADA v3.3.2, and the scripts were updated accordingly in v6.0.1 to read hazards in the new format. Additionally, I implemented minor fixes, such as updating the “cover” column in *FL_entity_Acelhuate_parametric.xlsx* and switching the base map in *plot_salvador_ma* from Stamen Terrain to OpenStreetMap Mapnik, as Stamen Terrain is no longer open source. The issue was formally reported in the CLIMADA-python repository ([link](#)), and a corresponding pull request ([link](#)) was submitted and successfully merged, providing a permanent fix. Beyond the technical fixes, the process of reporting the issue, proposing a solution, and successfully having my Pull Request merged into the official CLIMADA-python repository gave me firsthand experience in open-source collaboration.

2.3. Additional Task

The assets map of crops (Maize), livestock (Camel, Cattle, Goat, Sheep), rangeland, water sources, and Frankincense for Somalia. Data were downloaded from different sources, like the Food and Agriculture Organization (FAO), the ASAP system (Anomaly Hotspots of Agricultural Production), ESA WorldCereal, etc. I utilized my cartography and GIS skills to effectively visualize and present data using QGIS.

In addition, technical support was provided during the Training of Trainers for the Pakistan team, particularly in resolving issues related to CLIMADA installation.

3. Reflection

Working with the Analytics team at MCII was an incredible and unforgettable experience. I would like to express my deepest gratitude to my supervisor for being very supportive and flexible. This opportunity allowed me to significantly deepen my technical and analytical skills, including proficiency in Python programming, handling MATLAB and HDF5 file formats within the CLIMADA tool.

I gained valuable exposure to advanced concepts such as retrieval-augmented generation (RAG) architectures, along with an enriching experience in developing and deploying web applications leveraging RAG frameworks and their underlying large language models (LLMs). Beyond the technical skills, this experience enriched my understanding of climate risk assessment and climate risk adaptation strategies, providing a holistic view of how data-driven approaches can help in informed decision-making in the climate risk domain. Furthermore, I had the opportunity to observe how the United Nations system operates and how international collaborations function within UNU. Overall, this experience not only strengthened my technical capabilities but also broadened my perspective on global climate action and interdisciplinary collaboration.

I would recommend this internship to individuals who are keen on gaining hands-on experience in the Economics of Climate Adaptation (ECA), hazard modelling, and climate risk assessment. In addition, interns will have the chance to strengthen their Python programming skills in a similar domain. I would also recommend exploring the [CLIMADA-python](#) tool, an open-source platform for climate risk and adaptation modelling, and contributing to the project, as it is widely used within the UNU-EHS research community and beyond.

References

United Nations University. (n.d.-a). UNU Institute for Environment and Human Security. Retrieved September 4, 2025, from <https://unu.edu/ehs>

United Nations University. (n.d.-b). *Climate Finance & Insurance*. UNU-EHS. Retrieved September 4, 2025, from <https://unu.edu/ehs/our-work/risk-adaptation/climate-finance-insurance>

United Nations University (n.d.-c). *Climate Risk Analytics*. UNU-EHS. Retrieved September 4, 2025, from <https://unu.edu/ehs/our-work/risk-adaptation/climate-finance-insurance/climate-risk-analytics>

Amazon Web Services, Inc. (n.d.). *What is Retrieval-Augmented Generation?* Retrieved September 4, 2025, from <https://aws.amazon.com/what-is/retrieval-augmented-generation/>

RAG Application Demonstration Video Link:

<https://drive.google.com/drive/folders/13tAKrdPyoF3YFE5Y2n726u2VCIXrWKPK>